# Transfer: circuits and induction

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Apply circuit relationships and explain an induction observation.

**SUCCESS CRITERIA - I can...**

1. Use current and voltage rules in a new circuit.
2. Account for electrical power and energy.
3. Explain induction through changing flux.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “resistance” and “power” to explain your answer. Sentence starters are optional.

**resistance**: voltage divided by current for a stated operating condition

**power**: rate of energy transfer

**induction**: producing an emf through a changing magnetic flux

## 01 Warm-up: bring back an earlier idea

A motor is **62%** efficient and the following generator is **71%** efficient. **Find the combined useful fraction** in sequence.

**Word help:** generator = a device converting mechanical input to electrical output

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $V = IR$; $P = IV$; $E = P\Delta t$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$. Use stated facts about the circuit or coil. Ideal battery and wires unless stated otherwise.

**Worked reasoning:** A **12 V** supply drives a **6 ohm** resistor: current **2 A** and power **24 W**. A complete explanation checks both the charge-flow rate and the energy-transfer rate.

**Your turn:** 1. **Draw the circuit** and an energy-flow arrow.
2. **Label what an ammeter** and voltmeter would measure and where each connects.

**Sentence starter:** In my work, I used … to show … . I labeled it … .




Label component values, current and voltage as needed.

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

**Sketch two resistors** in parallel across a battery and **label the equal-voltage branches**.

**Sentence starter:** In my work, I used … to show … . I labeled it … .



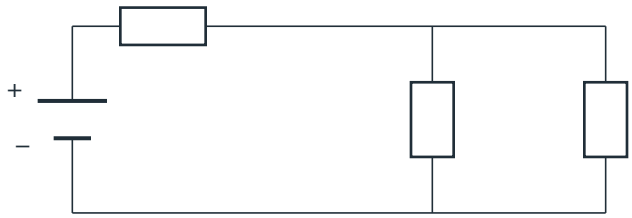

Circuit symbols: long/short lines = battery; rectangles = resistors.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A **12 V** battery powers a **4 Ω** resistor in series with two parallel **8 Ω** resistors.
1. **Label the network**.
2. **Calculate total current**, each branch current and total input power.

**Sentence starter:** I found … [unit]. This value means … in this situation.




**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Explain your reasoning

A magnet is held motionless near a fixed coil, then moved into it. **Explain the change** in the meter response without saying the magnet supplies unlimited energy.

**Sentence starter:** For this answer, I treated … as true. I found … . This supports … because … .

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

Measured (I, V) pairs are (**0.10**, **4**), (**0.20**, **8**), (**0.30**, **12**) in A, V. **Find resistance** and **explain how the measurements support** treating the resistance as constant.
**Draw the stated circuit** and **label quantities relevant** to the question.

**Word help:** ohmic = having constant resistance under the stated physical conditions

**Sentence starter:** For this answer, I treated … as true. I found … . This supports … because … .

Draw the complete circuit and label shared nodes.

**Marzano check - circle:** 1 2 3

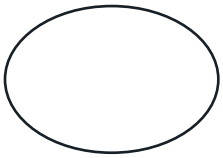
## 07 Exit check

A **50**-turn coil has flux per turn change by **0.004 Wb** in **0.20 s**. **Find average emf magnitude**.

**Word help:** magnitude = size of a quantity without its direction

**Sentence starter:** I found … [unit]. This value means … in this situation.

Label field direction, loop area and what changes.



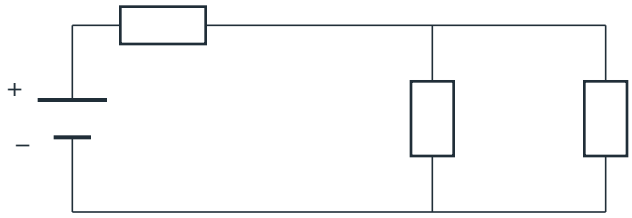
Turns: ____ Time interval: ____ emf: ____

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

A **12 V** source feeds a **2 ohm** series resistor and a parallel pair of **4 ohm** resistors. **Find total power** and power in each resistor; verify the energy-rate sum.

**Sentence starter:** I found … [unit]. This value means … in this situation.




**Marzano check - circle:** 1 2 3